ANALYSTLAB AFRICA CONSULTING

SwiftLog Logistics

Executive Business Intelligence Report

Supply chain analytics for the leadership team | Data: 1 January to 8 September 2026 | Prepared by **Emeh Franklin Chukwuebuka**, Data Analytics Track

Key Takeaways

- **Delivery is the biggest problem.** Only 36.28% of shipments arrive on time against an assumed 80% target, and the best warehouse (W01) reaches only 41.3%.
- **Only a third of orders are delivered.** 33.2% of orders are Delivered, 34.3% Pending and 32.5% Cancelled, yet every order has a shipment. Delivered revenue is ₦7.11M of ₦21.67M total order value.
- **Stock is large but badly spread.** 1,007,764 units are held against 82,442 ordered (12.2 times), yet 160 products fall below 100 units in at least one warehouse.
- **Revenue is widely spread.** The top 10 customers earn 5.05% and the top 10 products 9.85% of delivered revenue, although the top 200 customers carry 52.5%.
- **Returns are a process issue.** The return rate (9.46%) is almost equal across categories and 63% of returns are Wrong Item or Damaged.

1. Business Context and Objectives

SwiftLog operates warehouses, suppliers, inventory systems, shipment tracking and nationwide delivery. Rapid growth has brought delayed deliveries, inventory challenges, supplier performance issues and limited executive visibility. This report summarises a Power BI solution built to monitor performance, improve delivery efficiency, reduce operating costs, improve inventory and warehouse operations, evaluate suppliers, reduce returns and support decisions.

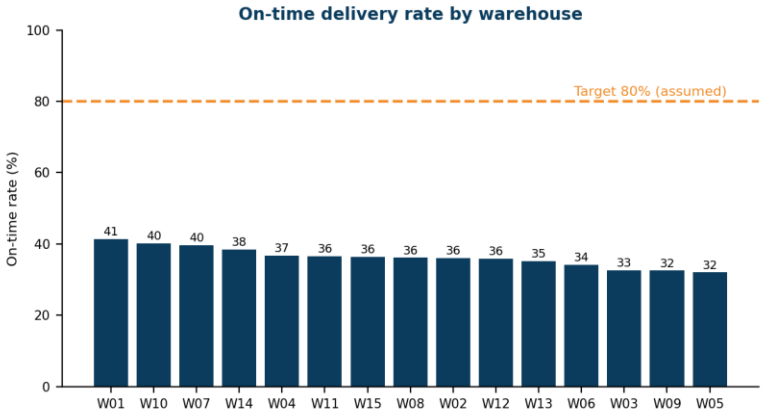
2. Approach

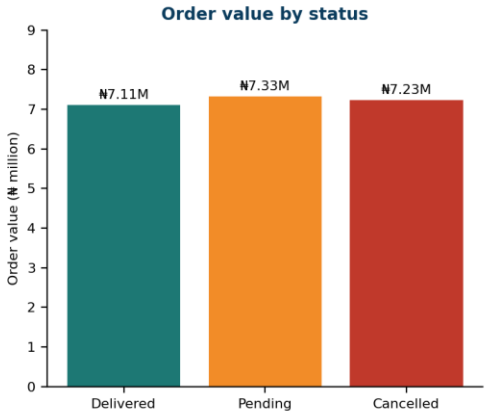
Eleven datasets (customers, orders, order items, products, shipments, delivery performance, returns, inventory, warehouses, employees and suppliers) were cleaned in Power Query and organised in a star schema of 5 dimension and 6 fact tables. DAX measures cover every required KPI, and a five-page interactive dashboard (Executive Overview, Customers, Inventory and Warehouse, Shipment and Delivery, Supplier and Product) presents the results. Assumptions are listed in section 5.

3. KPI Snapshot/Documentation

KPI	Value	What it means
Delivered revenue	₦7.11M	Quantity × unit price of Delivered orders (total order value ₦21.67M)
Orders / customers / products	5,000 / 1,000 / 250	991 customers have ordered; 5.05 orders per active customer
Average order value	₦4,504	Delivered revenue per delivered order with items
On-time delivery rate	36.28%	On Time = 0 to 2 days late; target of 80% is assumed
Average delivery delay	3.57 days	5.03 days among delayed shipments
Cancellation rate	32.52%	1,626 cancelled orders worth ₦7.23M
Return rate	9.46%	473 returned orders; 500 returns
Customer retention	95.96%	Customers with 2 or more orders (951 of 991)

KPI	Value	What it means
Inventory stock level	1,007,764 units	Sum of stock movements across 15 warehouses
Inventory turnover	0.027	Units delivered divided by units in stock
Warehouse utilisation (est.)	83.98%	Stock ÷ an assumed 80,000-unit capacity per warehouse
Supplier performance score	58.3 of 100	Network-level framework (supplier data not linked)





4. Key Findings

- **Delivery.** 3,186 shipments (63.7%) are delayed and 1,965 of them are 5 to 7 days late. Delays are company-wide: the best and worst warehouses differ by only 9 points (W01 41.3%, W05 32.0%), and workload and stock level show almost no link to on-time rate. By region, Kano customers fare worst (34.3% on time) and Lagos best (39.0%); Lagos warehouses ship 53% of all shipments.
- **Order completion.** All 3,342 Cancelled and Pending orders still have a shipment, and 315 returned orders are Cancelled or Pending. This points to wasted shipping cost and unreliable order status.
- **Inventory and warehouses.** Turnover is 0.027. Estimated utilisation runs from 71% (W01) to 97% (W12). Shipments per employee range from 19.9 (W13) to 37.8 (W14), so staffing is uneven.
- **Customers and products.** Revenue by city is close (Kano ₦1.86M, Lagos ₦1.81M, Ibadan ₦1.76M, Abuja ₦1.68M). Fashion leads categories (₦2.73M, 38.4%), and the top 10 products are priced ₦416 to ₦495. Ibadan has 262 customers (26%) but no warehouse; 52% of its shipments leave from Lagos.
- **Returns.** Late Delivery 185, Wrong Item 163, Damaged 152. Returned delivered orders carry ₦715,766 (10.1% of delivered revenue). Late Delivery returns are no more delayed than others, so the reason code may be unreliable.
- **Peak months.** August is the peak (revenue ₦997,021; 654 orders; lowest cancellation rate 29.2%). July is the weakest revenue month (₦741,706). September covers only 8 days.
- **Suppliers.** The data holds 150 suppliers but nothing links them to products or shipments, so supplier performance cannot be ranked.

5. Assumptions and Data Limitations

- **Data:** 11 files covering 1 January to 8 September 2026; no missing values or broken links were found.

- **On Time vs Delayed:** inferred from the data (0 to 2 days vs 3 to 7 days). The 80% on-time target is assumed.
- **Revenue:** quantity × unit price for Delivered orders; no discounts, costs or tax; currency not stated (naira used).
- **Order status** conflicts with shipments (Cancelled and Pending orders have shipments and returns); status is kept separate from delivery KPIs.
- **Inventory** is the sum of stock movements per warehouse and product (as advised), and may overstate current stock. Utilisation assumes 80,000 units of capacity per warehouse and is labelled estimated.
- **Returns** link to orders, not products, so product return rates are approximate and small samples can show high rates by chance.
- **Suppliers** are a standalone table; no supplier ranking is claimed. **No delivery timestamps:** delivery time is estimated from delay days.

Recommendations

Twelve actions, each tied to a finding from the dashboard. Impact figures are illustrative arithmetic on revenue, not profit forecasts, and the targets are proposals.

S/N	Recommendation	Evidence	Expected impact
1	Set an interim 50% on-time target, escalate shipments over 2 days late, and copy the practices of W01, W10 and W07.	On-time rate 36.28%; 1,965 shipments 5 to 7 days late	+10 points = about 500 more on-time shipments
2	Dispatch only confirmed orders, clear the 1,716 pending orders and align order status with shipments.	3,342 cancelled or pending orders all shipped	Fewer wasted shipments; 10 points fewer cancellations = about 500 orders (₦2.2M)
3	Set maximum stock levels and reorder points, and run a physical stock count.	Stock is 12.2 times units ordered; turnover 0.027	Frees working capital and space
4	Redistribute stock to the 160 products that run short in a warehouse.	264 warehouse-product lines under 100 units	Fewer stock-outs without buying more
5	Rebalance staffing so no warehouse exceeds about 30 shipments per employee.	19.9 to 37.8 shipments per employee	More even workload
6	Add pick-and-pack checks and stronger packaging; verify Late Delivery reason codes.	Wrong Item and Damaged are 63% of returns; ₦715,766 returned	Return rate toward 7%
7	Offer a loyalty programme to the top 200 customers and a win-back offer to 132 low-frequency customers.	Top 20% of customers earn 52.5% of revenue	Protects half of delivered revenue
8	Add SupplierID to products and shipments, then score suppliers on delivery, returns and lead time.	150 suppliers, no link; network score 58.3	Enables supplier rankings and service agreements
9	Pilot nearest-warehouse routing for Ibadan and Kano and assess an Ibadan hub.	Only 24.9% of shipments leave from the customer's city	Shorter routes if the pilot succeeds

S/N	Recommendation	Evidence	Expected impact
10	Run a weekly KPI scorecard with targets, and add delivery timestamps and warehouse capacity to the data.	Delivery dates and capacity are missing	Clear accountability for each KPI
11	Expand the high-priced Fashion and Food range and promote it in weaker months.	Fashion 38.4% of revenue; July weakest (₦741,706)	More revenue from existing customers
12	Plan stock and staffing ahead of the August peak.	August: ₦997,021 and 654 orders	Smoother service at peak